

BILAL ELKHOULY

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Education

University of Guelph

Dec 2025

Bachelor of Engineering, Computer Engineering; Current GPA: 85.45 (3.76)

Guelph, ON

- College of Engineering & Physical Sciences Dean's Honours List [Fall 2021, Winter 2022, Fall 2023, Winter 2023, Winter 2024, Fall 2024, Winter 2025]

Work Experience

University of Guelph

May 2025 – Aug 2025

Software Developer

Guelph, ON

- Developed and deployed full-stack **React + Django** application by integrating APIs, user input forms, and backend model logic, which improved data processing speed and enhanced frontend responsiveness.
- Set up and integrated **Amazon RDS (PostgreSQL)** with Django backend, ensuring secure database connectivity and enabling reliable persistent storage for multi-user production environments.
- Built and maintained automated **CI/CD pipelines** with GitHub Actions by enforcing test coverage and pull request checks, which streamlined deployments to **AWS Amplify** and **Elastic Beanstalk** and reduced integration errors.
- Collaborated in an agile development team to deliver user-facing features and publication tools across **React and WordPress**, resulting in improved usability and accessibility for internal stakeholders.
- Contributed to the development of **Brisk**, a machine learning experimentation framework, by implementing CLI utilities, run-document snapshotting, and config file generation, which enhanced monitoring, reproducibility, and workflow automation.

University of Guelph

Sep 2024 – April 2025

Software Programmer

Guelph, ON

- Developed and deployed scalable backend systems using Python and MySQL, optimizing journal hosting services for Open Journal Systems (OJS).
- Developed a **custom author review form plugin** for Open Journal Systems (OJS), enabling a new workflow where authors could directly respond to peer reviewers' pre-filled comments, streamlining communication, enhancing feedback clarity, and reducing administrative overhead by 25%.
- Deployed and configured **journal hosting on a cPanel server** using **MySQL**, ensuring secure management of submission records, which improved data retrieval times by 20%.
- Monitored and optimized server performance through regular maintenance, enhancing user experience for journal editors and reviewers.
- Collaborated with faculty to align technical implementations with academic objectives, refining OJS workflows to streamline the peer-review process.

University of Guelph

Sep 2022 – April 2025

Engineering Peer Helper

Guelph, ON

- Overhauled the department's website using content management systems (CMS), enhancing usability and ensuring up-to-date, informative content.
- Facilitated interactive class sessions and problem-solving activities for a group of 30+ engineering students, enhancing their understanding of key concepts.

University of Guelph

Sep 2024 – Dec 2024

*Undergraduate Teaching Assistant (UTA) for ENGG*1410*

Guelph, ON

- Guided **over 50** first-year engineering students in mastering **C programming concepts** during lab sessions, resulting in an 85% average lab completion rate.
- Provided one-on-one support to students, effectively clarifying complex topics and troubleshooting coding issues.
- Collaborated with faculty to optimize lab materials and schedules, reducing session delays and increasing student engagement.

Projects

Quoma

Next.js — JavaScript — Supabase

- Built a fully functional **Next.js frontend** with **Tailwind CSS** and **Shadcn**, enabling users to create and manage automated workflows through a visual flowchart interface, simplifying research, responses, and outreach processes for users
- Implemented real-time data management using **Supabase**, incorporating autosaving and CRUD operations, ensuring consistent data integrity and reducing data recovery errors by 20%.
- Optimized backend integrations, reducing API response times by 40%, which improved user satisfaction and overall system performance

Merchandise Route Tracker

Next.js — Node.js — AWS

- Developed a **Next.js frontend** that allowed companies to track their merchandisers' real-time locations via **AWS API Gateway**, increasing operational transparency and reducing unvisited delivery points by 15%.
- Created **Node.js code** for **AWS Lambda functions** to ensure continuous tracking and logging of merchandiser locations, reducing data discrepancies and saving storage space by 25% through efficient log management in **AWS S3**.

Certifications & Skills

- **Languages:** Python, Java, PHP, SQL, HTML5, CSS3, JavaScript, C, VHDL, MIPS Assembly
- **Developer Tools:** Git, Docker, Postman, Jira, IntelliJ Idea, Agile Software Development Process, Xilinx Vivado
- **Hardware Design Skills:** Digital Logic Design, FPGA Programming, Microcontroller Interfacing, and Embedded System Development
- **Libraries/Frameworks:** React.js, Next.js, Node.js, Supabase, MongoDB, PostgreSQL, MySQL, AWS, Content Management Systems (CMS), Pytorch, Tensorflow
- **Certifications:** Solidworks CSWA, 100 Days of Code: The Complete Python Pro Bootcamp